

Lyapunov exponent for discrete-states and discrete-time systems

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1 Preliminaries

Let $G(V, E)$ denote a network with N nodes (or cells), $V = \vec{x} = \{x_1, x_2, x_3, \dots, x_N\}$ and $|E|$ edges. Let $\mathbb{S} = \{0, 1\}$ denote the set of all possible states of a cell. Thus, the configuration space of the network is $\mathbb{S}^N$. Consider $\Omega(x_i)$ as the neighborhood set of the node x_i . The transition function, denoted by $\Phi : \mathbb{S}^{|\Omega|} \rightarrow \mathbb{S}$, specifies that the state transition of node x_i is influenced by its neighborhood. This transition function Φ establishes a global mapping $\mathbf{S} : \mathbb{S}^N \rightarrow \mathbb{S}^N$, where $s_i(x_i, t) = \Phi(\Omega(x_i))$, and $\mathbf{S} = (s_1, s_2, \dots, s_N)$ is the vector state. We can denote the initial vector state as $\mathbf{S}^0 = (s_1(x_1, 0), s_2(x_2, 0), \dots, s_N(x_N, 0))$ and the vector state at step t as $\mathbf{S}^t = (s_1(x_1, t), s_2(x_2, t), \dots, s_N(x_N, t))$.

2 Boolean derivatives and the Jacobian

Given the definitions above, we can define the boolean derivative.

$$\frac{\partial s_i}{\partial x_j} = \begin{cases} s_i(x_1, x_2, \dots, x_j \oplus 1, \dots, x_N) \oplus s_i(x_1, x_2, \dots, x_N) & \text{if a changing in } x_j \text{ cause a transition in } s_i \text{ at step } t+1 \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

where $\oplus$ is the XOR operator (module-2 summation)[1]. The boolean derivative will be 1 if and only if a changing in the x_j state at time t (i.e. s_j^t) can induces a transition in the state of x_i at time $t + 1$ (i.e. changing in s_i^{t+1}).

Let be $\mathbf{s}^0$ the initial state of all cells and $\mathbf{s}^t$ the state at time t , then the Jacobian is defined by:

$$J(\vec{x}^t, t) = \begin{bmatrix} \frac{\partial s_1(x_1, t+1)}{\partial x_1} \Big|_t & \frac{\partial s_1(x_1, t+1)}{\partial x_2} \Big|_t & \dots & \frac{\partial s_1(x_1, t+1)}{\partial x_N} \Big|_t \\ \frac{\partial s_2(x_2, t+1)}{\partial x_1} \Big|_t & \frac{\partial s_2(x_2, t+1)}{\partial x_2} \Big|_t & \dots & \frac{\partial s_2(x_2, t+1)}{\partial x_N} \Big|_t \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial s_N(x_N, t+1)}{\partial x_1} \Big|_t & \frac{\partial s_N(x_N, t+1)}{\partial x_2} \Big|_t & \dots & \frac{\partial s_N(x_N, t+1)}{\partial x_N} \Big|_t \end{bmatrix} \quad (2)$$

The Jacobian describes how n mutually orthogonal perturbation propagate from the time t to time $t + 1$.

3 Configuration space

The configuration space represents all nodes and their respective states within a network. In the context of a network, this space is represented as $\mathbb{S}^N$, constituting an N -dimensional domain.

Consider $\vec{x}^0$ as the initial configuration. A perturbation, or defect, denoted by $\delta \vec{x}^0$, induces a deviation in the trajectory within the configuration space over a time period T . Let $\Delta \vec{x}^0$ the difference pattern between the original trajectory on configuration space and the disturbed trajectory at step zero. This value can be measured by:

$$\Delta \vec{x}^0 = \mathbf{S}^0(\vec{x}^0) \oplus \mathbf{S}^0(\vec{x}^0 \oplus \delta \vec{x}^0) \quad (3)$$

Therefore, $\Delta \vec{x}^0$ measures the displacement of the trajectories caused by a perturbation at the initial time. We can investigate whether such perturbations lead to convergence or divergence in the trajectories within the configuration space evolving them by the jacobian.

$$\Delta \vec{x}^{t+1} = J(\vec{x}^t, t) \otimes \Delta \vec{x}^t \quad (4)$$

Equation 4 operates recursively, meaning that to determine the difference pattern at time $t + 1$, we first calculate the difference pattern at time t . In turn, computing the difference pattern at time t requires considering the difference pattern at time $t - 1$, and so forth. Here, the operator $\otimes$ denotes standard matrix multiplication, with the addition operation substituted by $\oplus$.

Convergence of $\Delta \vec{x}^{t+1}$ means stability, while exponential divergence suggests chaotic behavior.

3.1 The Lyapunov exponent

In the configuration space, the Lyapunov exponent quantifies the rate of growth of the difference pattern between the original trajectory and the perturbed trajectory. The Lyapunov exponent, as introduced in [2], is defined as follows:

$$\lambda(T) = \frac{1}{T} \sum_{t=0}^T \log\left(\frac{|\Delta \vec{s}^{t+1}|}{|\Delta \vec{s}^t|}\right) \quad (5)$$

where $\Delta \vec{x}^{t+1} = \sum_{i=1}^N \Delta x_i^{t+1}$

4 Tangent space

The tangent space of a CA is a conceptual space that allows for the analysis of how small perturbations or defects evolve in the system. It is analogous to the tangent space in continuous-state dynamical systems, where it represents the space of all possible directions in which perturbations can grow or shrink. This space is essentially a vectorial space, delineated by the tangent vector at a point $P \in \mathbb{S}^N$, and it exhibits an isomorphism with the space $\mathbb{S}^N$. The advantage of working with perturbations in tangent space is that we only need to keep track of the evolution of n mutually orthogonal perturbations.

The tangent space has the linearity property, it means that we can decompose the perturbation vector into a complete and orthogonal set of vectors representing the perturbation in each direction: $\delta \vec{x}^t = \sum_{i=1}^N \delta x_i^t \vec{e}_i$, where $\{\vec{e}_i\}$ denotes the standard basis. These components evolve in accordance with the Jacobian:

$$\begin{bmatrix} \delta x_1^{t+1} \\ \delta x_2^{t+1} \\ \vdots \\ \delta x_N^{t+1} \end{bmatrix} = \begin{bmatrix} \frac{\partial s_1}{\partial x_1}|_t & \frac{\partial s_1}{\partial x_2}|_t & \cdots & \frac{\partial s_1}{\partial x_N}|_t \\ \frac{\partial s_2}{\partial x_1}|_t & \frac{\partial s_2}{\partial x_2}|_t & \cdots & \frac{\partial s_2}{\partial x_N}|_t \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial s_N}{\partial x_1}|_t & \frac{\partial s_N}{\partial x_2}|_t & \cdots & \frac{\partial s_N}{\partial x_N}|_t \end{bmatrix} \begin{bmatrix} \delta x_1^t \\ \delta x_2^t \\ \vdots \\ \delta x_N^t \end{bmatrix} \Leftrightarrow \delta \vec{x}^{t+1} = J(\vec{x}^t, t) \delta \vec{x}^t \quad (6)$$

Being the tangent space linear, and decomposing the perturbation vector by a sum of its orthonormal components, we can build a matrix Y^0 which represents all orthogonal perturbations that form the basis of the tangent space and, given such orthogonality, this matrix is the identity matrix. This matrix symbolize a sphere of perturbation emanating from the initial state.

The temporal evolution, as well as in the configuration space, is given by the Jacobian, but instead of evolving the difference pattern vector, we will evolve the matrix Y^t from Y^0

$$\mathbf{Y}^{t+1} = J(\vec{x}^t, t) \mathbf{Y}^t = J(\vec{x}^t, t) \cdots J(\vec{x}^0, t) \mathbf{Y}^0 \quad (7)$$

We can diagonalize the matrix Y defining the Γ matrix, which is: $\Gamma = Y \cdot (Y)^{\text{transpose}}$.

To calculate the Lyapunov exponent within the tangent space, we extract the eigenvalues Λ_i^t of Γ^t , satisfying the equation $\mathbf{Y}^t v_i = \Lambda_i^t v_i$, where v_i denotes the i -th eigenvector of Γ^t . The Lyapunov spectrum is defined as follows:

$$\lambda_i = \lim_{t \rightarrow \infty} \frac{1}{t} \ln(\Lambda_i^t) \quad (8)$$

where λ_i is the i -th Lyapunov exponent.

We can further define the total Lyapunov exponent, which quantifies the rate of growth or decay of solutions within a dynamical system. It plays a crucial role in examining whether trajectories close to each other within the system tend to converge towards or diverge from a specified equilibrium point. The total Lyapunov exponent is characterized by:

$$L = \lim_{t \rightarrow \infty} \frac{1}{t} \log(\delta \vec{s}^t) \quad (9)$$

The Lyapunov profile provides a normalized logarithmic representation of the count of perturbed paths reaching each cell, all originating from a single initial perturbation.

5 Relationship between damage spreading and the network diameter

Considering a initial perturbation $\delta\vec{x}^0$ upon a subset of nodes, at step t , the nodes affected by perturbations form a subnetwork $G_{\Delta}^t \subseteq G$. Utilizing G_{Δ}^t , we can determine the rate of spread of the variation pattern, denoted by γ to a cellular automaton model on the network in the following manner:

$$\gamma(t) = \frac{1}{t}D(G_{\Delta}^t) \quad (10)$$

where $D(G_{\Delta}^t)$ represents the diameter of the subnetwork affected by the spreading of perturbations.

6 Interpretation of both formalisms

In configuration space, the Lyapunov exponent indicates how quickly small initial differences between two states will grow. A positive Lyapunov exponent suggests that small perturbations will lead to significant divergence over time, indicating chaotic behavior. Conversely, a negative exponent implies that perturbations will die out, leading to stable behavior.

In tangent space, the Lyapunov exponent provides a more detailed picture of the system's stability by quantifying the exponential rates at which small perturbations in various directions grow or shrink. This approach offers a finer resolution of the system's dynamics, capturing the behavior of individual perturbations and their interactions over time. The spectrum of Lyapunov exponents from tangent space analysis can reveal the presence of chaotic dynamics even if the configuration space does not exhibit exponential growth directly.

In summary, while both spaces offer insights into the stability and dynamics of CA, the configuration space gives a more global view of the system's behavior under perturbations, and the tangent space provides a detailed, linearized analysis of how these perturbations evolve. The Lyapunov exponents in tangent space, being derived from a linearized model, can reveal intricate details about the system's stability and chaotic nature that might not be apparent from the configuration space alone.

References

- [1] J. M. Baetens, B. De Baets, On the topological sensitivity of cellular automata, *Chaos: An Interdisciplinary Journal of Nonlinear Science* 21 (2) (2011).
- [2] B. Luque, R. V. Solé, Lyapunov exponents in random boolean networks, *Physica A: Statistical Mechanics and its Applications* 284 (1-4) (2000) 33–45.